

Experiment 1: Design Zero-shot, One-shot, and Few-shot Prompts to Generate a Product Description for a Smart Fitness Watch

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch.

Python Code:

```
from openai import OpenAI

# Replace with your NEW OpenAI API Key
client = OpenAI(
    api_key="sk-proj-oXkP1XbG7lavQbYlgQ-_zC8PQ7rY-
qLYwkq2HVthAKl40D4LbWGzCy6NYOv6ogGhgxo70ls0grT3BlbkFJOYc7uN4moo5M3-
luPCn-H7JOEfoTuHmha15XBexnkMM-aUCd0UPYSthtYb0Nmz4yzwAiBGEPsA"
)

# ----- Zero-shot Prompt -----
prompt = "Generate a product description for a Smart Fitness Watch."

response = client.responses.create(
    model="gpt-5.4-mini",
    input=prompt
)

print("===== ZERO-SHOT PROMPT =====")
print(prompt)
print("\n===== ZERO-SHOT OUTPUT =====")
print(response.output_text)

# ----- One-shot Prompt -----
prompt = """
Example:

Product: Wireless Bluetooth Earbuds

Description:
Wireless Bluetooth Earbuds provide crystal-clear sound, noise cancellation,
touch controls, and up to 24 hours of battery life. They are lightweight,
comfortable, and perfect for music and calls.

Now generate a similar product description for a Smart Fitness Watch.
```

```
"""
```

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=prompt  
)
```

```
print("\n\n===== ONE-SHOT PROMPT =====")  
print(prompt)  
print("\n===== ONE-SHOT OUTPUT =====")  
print(response.output_text)
```

```
# ----- Few-shot Prompt -----
```

```
prompt = """  
Example 1
```

Product: Wireless Mouse

Description:

A wireless mouse with ergonomic design, adjustable DPI, smooth tracking, and long battery life.

Example 2

Product: Smart Speaker

Description:

A smart speaker with voice assistant support, high-quality sound, Wi-Fi connectivity, and smart home control.

Now generate a similar product description for a Smart Fitness Watch.

```
"""
```

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=prompt  
)
```

```
print("\n\n===== FEW-SHOT PROMPT =====")  
print(prompt)  
print("\n===== FEW-SHOT OUTPUT =====")  
print(response.output_text)
```

Output:

===== ZERO-SHOT PROMPT =====

Generate a product description for a Smart Fitness Watch.

===== ZERO-SHOT OUTPUT =====

Meet the Smart Fitness Watch—your all-in-one companion for a healthier, more connected lifestyle. Designed to keep up with your day, it tracks your steps, heart rate, calories burned, sleep, and workouts with precision, helping you stay on top of your fitness goals.

With a sleek, lightweight design and a bright, easy-to-read display, this watch looks great whether you're at the gym, in the office, or on the go. Receive call, text, and app notifications right on your wrist, so you never miss an important update. Plus, with long battery life and water-resistant construction, it's built to handle your busiest days and toughest workouts.

Whether you're starting your fitness journey or pushing toward your next milestone, the Smart Fitness Watch gives you the insights, motivation, and convenience you need to move smarter every day.

===== ONE-SHOT PROMPT =====

Example:

Product: Wireless Bluetooth Earbuds

Description:

Wireless Bluetooth Earbuds provide crystal-clear sound, noise cancellation, touch controls, and up to 24 hours of battery life. They are lightweight, comfortable, and perfect for music and calls.

Now generate a similar product description for a Smart Fitness Watch.

===== ONE-SHOT OUTPUT =====

****Product:**** Smart Fitness Watch

****Description:****

The Smart Fitness Watch helps you track your health and stay connected throughout the day. It monitors your heart rate, steps, calories burned, sleep quality, and workouts with ease. Featuring a sleek design, customizable watch faces, smart notifications, and water resistance, it's comfortable to wear all day and perfect for fitness and everyday use.

===== FEW-SHOT PROMPT =====

Example 1

Product: Wireless Mouse

Description:

A wireless mouse with ergonomic design, adjustable DPI, smooth tracking, and long battery life.

Example 2

Product: Smart Speaker

Description:

A smart speaker with voice assistant support, high-quality sound, Wi-Fi connectivity, and smart home control.

Now generate a similar product description for a Smart Fitness Watch.

===== FEW-SHOT OUTPUT =====

Product: Smart Fitness Watch

Description:

A smart fitness watch with heart rate monitoring, activity tracking, sleep analysis, GPS support, water resistance, and smartphone notifications.

Result

Few-shot prompting generated a more focused and structured product description compared to Zero-shot and One-shot prompting